

MICHAEL ZAUGG

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U.S Citizen | Active Security Clearance:T3 (Secret)

Computer Engineering student and active Secret-cleared intern specializing in embedded systems, firmware development, and hardware design. Experienced in developing C/C++ and Python applications for STM32-based systems, designing and debugging PCBs, and integrating sensors, RFID, displays, and power distribution hardware. Proven experience leading technical teams through hardware reverse engineering, legacy system modernization, and open-source development, with hands-on experience in FreeRTOS, KiCad, and CAD. Experienced in taking embedded systems from electrical and mechanical design through software integration, testing, and validation.

Education/Certificates

Brigham Young University - Idaho

Bachelor of Science, Computer Engineering

GPA: 3.66

Dec 2026 (Expected)

Rexburg, ID

Embedded Systems Certificate

Brigham Young University - Idaho

April 2026

330 Completed Hours

Qualifications

Equipment: Oscilloscope, Digital Multimeter, LCR Meter, Function Generator

Electronics: Arduino, STM32, Op-Amps, BJTs, ADCs, ESP32, RaspberryPi

Programming: C, C++, C#, Python, System Verilog, FreeRTOS

Software: KiCad, LTSpice, Logisim, Visual Studio Code, Microsoft Office, Altium

Team: Team Leadership, Agile/Scrum, Technical Mentorship

Projects

Hill AFB 309 SWEG - Power Supply CCA (Capstone Project)

Sept 2026 - Dec 2026

Senior Group Project | ~13hrs/week | Student

- Collaborated with a small team to modernize legacy SDS Circuit Card Assembly (CCA) hardware originally designed in the 1990s.
- Redesigned electrical schematics and PCB layouts to meet Class 2 or higher IPC standards and improve manufacturability.
- Researched and selected replacement components with End-of-Life (EOL) dates more than 10 years in the future to improve long-term system maintainability.
- Generated updated system requirements, Gerber files, schematics, and Bills of Materials (BOM) for Hill AFB.
- Developed and documented a test, validation, and verification plan to evaluate the redesigned CCA against updated requirements.

Gambit - Embedded Systems Chess Game (Hill AFB, 309 SWEG)

Jun. 2026 - Present

Project Developer | 40hrs/week | NH-1

- Engineered a Python framework for e-Paper display integration to establish a low power, high contrast visual display interface for the system.
- Co-designed a dedicated power distribution Printed Circuit Board (PCB) alongside a team of engineers to supply clean, regulated power across all system microcontrollers.
- Developed C++ drivers for a 64-antenna array to enable real time Radio Frequency Identification (RFID) tracking of physical chess piece movements across the board.
- Integrated the Stockfish AI chess engine into the embedded system architecture using Python, establishing native, low latency AI opponent capabilities.
- Designed an intuitive Python user interface to facilitate seamless user interactions and game customization against the internal AI engine.

- Collaborated on physical enclosure designs using Computer Aided Design (CAD) software to ensure fitment for the PCB, sensors, and display hardware.

- Reversed Engineered Chinese RFID reader circuits for use with chess board PCB design.

Reversed Engineered Controller (Personal Project)

Oct. 2024 - Sept 2026

Project Leader | ~5hrs/week | Volunteer

<https://github.com/MichaelZaugg/OpenSteamController-Continued/releases>

- Led a remote 7-member open-source team in analyzing and redesigning legacy hardware by providing technical guidance on PCB design and debugging to keep the community project moving forward.

- Executed schematic capture, electrical rules validation, and board layout using KiCad design software to ensure design reliability and accurate circuit replication.

- Designed and prototyped custom Flexible Flat Cable (FFC) replacement cables for trackpad and haptic components using low-level circuit analysis techniques to restore physical hardware functionality.

- Modeled 3D interior components and mounting structures using CAD software to verify mechanical tolerances and ensure custom replacement PCBs fit precisely inside original controller housings.

Artillery System Project (Embedded Systems Course)

Sept 2025 - Dec 2025

Semester Long Group Project | ~8hrs/week | Student

- Programmed STM32 stepper motor drivers in C to achieve precise control over turret rotation and firing mechanics.

- Engineered physics calculation algorithms and integrated Time-of-Flight sensors in collaboration with a 4-person team to maximize target acquisition accuracy and automated firing capability.

- Implemented concurrent FreeRTOS tasks for real-time task management across target acquisition, firing sequences, and user inputs to prevent control lag.

- Modeled and fabricated the custom turret mounting system using OnShape CAD to structurally support physical components and operational motor forces.

Brief Work History

Computer Scientist Intern

June 2026 - Present

309th Software Engineering Group (SWEG) - Hill AFB | 40hrs | NH-1

Hill, AFB

- Contributed to the end to end development of embedded defense applications, working within an agile team framework to take system requirements from initial concept to physical prototype and testing.

- Took technical and operational team training, executing software development best practices aligned with Air Force civilian engineering standards.

Fundamental Digital Systems TA

Sept 2025 - Dec 2025

Brigham Young University - Idaho | 20hrs | \$10.5/hr

Rexburg, ID

- Tutored students in digital logic concepts and reviewed project designs to increase comprehension and lab success.

- Partnered with faculty on weekly lab plans to test and implement new hands-on learning exercises.

Early Morning Custodian - Shift Lead

Sept 2023 - July 2025

Brigham Young University - Idaho | 24hrs | \$11/hr

Rexburg, ID

- Led shift operations and delegated weekly tasks to ensure high-priority areas met facility standards on schedule.

Additional Qualifications:

- Valid current U.S Passport
- Valid current Oregon Drivers License
- Enrolled in the Selective Service System
- Eagle Scout | ~2,500 hrs
- Student Pilot | ~200 hrs
- Civil Air Patrol | ~2,000 hrs
- Volunteer work in Brazil/ Utah | ~17,500 hrs | 2019-2021